LIFELEGENDS 3D • ENTERPRISE SPECIFICATION

COMPLETE PLATFORM AUDIT & COMMERCIAL FEATURE DOSSIER (¹50,00,000 / ,000 T

ARCHITECTURE PRINCIPLE: 100% FREE FOREVER (ZERO RUNNING COSTS, ZERO PAID APIS)

Repository: https://github.com/aetherpixels/LIFELEGEND.git | Next.js 16.4 Turbopack | TypeScript | Three.js

I. 3D WebGL Spatial Computing & Interactive Visuals

1. Interactive 3D Celestial Core (/)

Real-time Three.js interactive particle core with gyroscopic drag-and-rotate physics, dynamic lighting, and cinematic motion blur.

2. Interactive 3D Constellation Galaxy Map (/constellation)

Interactive WebGL star cluster mapping historical thinkers with 3D influence connection vectors, raycast orbital focus, and holographic dossiers.

3. Interactive 3D Historical Relic Viewport (<Artifact3DViewer />)

Embedded 3D inspection viewport inside biographies with live wireframe toggle, 360-degree auto-rotation, and orbit controls (Tesla Coil, Da Vinci Polyhedron, Curie Core).

II. AI & Conversational Intelligence Engines (Zero-Cost Client-Side)

4. AI Oracle Dialogue Engine (/oracle)

Real-time conversational simulation engine with historical personas (Tesla, Lovelace, Curie, Da Vinci, Einstein) using client-side semantic matching without third-party API bills.

5. AI Audio Narrator (<AudioNarrator />)

Browser-native neural voice synthesis reading full biography texts aloud with play, pause, pitch, and speed controls (0.8x-1.5x) with zero audio hosting fees.

III. Analytical & Comparative Research Tools

6. Titan Mind Comparison Matrix (/compare)

Side-by-side analytical matrix breaking down great thinkers across Era, Discipline, Innovation, Rivalry, Legacy, and Philosophy.

7. Personal Study Codex Vault (/codex)

Private research notebook with instant client-side persistence and one-click formatted research PDF dossier generation.

8. Interactive Historical Timeline Explorer (<HistoricalTimeline />)

Chronological life milestones with interactive era jumps and progress trackers.

9. Interactive Trivia Challenge Engine (<HistoricalTriviaQuiz />)

Gamified knowledge quiz dynamically tuned to each historical figure with live scoring and streak feedback.

10. Daily Wisdom & Quote Card Generator (<DailyQuoteGenerator />)

Rotating philosophical quote engine with instant clipboard copying and shareable visual quote cards.

IV. Global Navigation & Power-User Controls

11. Enterprise Command Palette (<CommandPalette /> — Cmd+K / Ctrl+K)

Global search and route-switcher modal supporting instant keyboard navigation across all historical figures, tools, and pages.

12. Unified Glassmorphism Navigation (<Navbar />)

Responsive header and mobile drawer with direct portals: Home, Lives, Constellation, Oracle, Compare, and Codex, plus user status indicator.

13. Dynamic Hero Exploration Hub (<HeroContent />)

High-impact landing hero with animated typography and direct portal chips to all flagship interactive tools.

V. Reading Enhancements, Gated Downloads & Monetization

14. Biography Reading Enhancements HUD (<BiographyReadingEnhancements />)

Live word-count reading time estimator, multi-step font resizing for accessibility, and Focus Mode.

15. Authenticated Gated PDF Download System

High-resolution biography PDF download portal requiring free account access to boost user retention.

16. Non-Intrusive Monetization Slot (<AdBanner />)
Clean neon-framed Google AdSense and sponsor banner placeholders for passive ad monetization.

VI. User Accounts, Content Management & Audio Architecture

17. Zero-Cost Client-Side Authentication Engine (AuthContext & <AuthModal />)
Frictionless modal authentication and local credential persistence with zero third-party subscription fees.

18. Headless CMS via Embedded Sanity Studio (/studio)
Full-featured CMS dashboard running on Next.js for real-time publishing, asset management, and PortableText editing.

19. Procedural Web Audio Engine (soundfx.ts)
Pure browser-synthesized audio effects (click ticks, hover blips, success fanfare) requiring zero MP3/WAV asset downloads.

20. Cosmic Ambient Space Synthesizer (<SpaceAmbientAudio />)
Dual-oscillator synthetic space drone with user volume controls and persistent state.

VII. Enterprise Security & Infrastructure Verification

21. Production Security Headers & Strict CSP (next.config.ts)
Configured with Content Security Policy, X-Frame-Options: DENY, X-Content-Type-Options: nosniff, Strict-Transport-Security (HSTS), and Permissions-Policy.

22. Input Sanitization & Secret Isolation
Client-side XSS sanitization across inputs and full environment variable isolation with verified clean git history.

23. Next.js 16.4.0 Turbopack Production Verification
Compiled and validated across all 11 routes with 0 errors and deployed to GitHub remote.